The use of the polymerase chain reaction (PCR) for clinical diagnosis



Modified from:

"The power of a test: How COVID-19 is diagnosed and who does it", by Alison Kieffer, Emaly J. Piecuch, Christina Vallianatos, & Sarah A. Wojiski. Genomic Education. The Jackson Laboratory, Bar Harbor, ME.

Corum, J. and C. Zimmer. 2020. How coronavirus hijacks your cells. The New York Times. <https://www.nytimes.com/interactive/2020/03/11/science/how-coronavirus-hijacks-your-cells.html>

In early 2020, Marcus stepped out of his car and into the sunshine, ready to head into the hospital for another day of work in the molecular diagnostics laboratory. A molecular diagnostics laboratory is just one of several types of laboratories that exist in hospitals. These laboratories are also referred to as medical laboratories, or clinical laboratories. The term "clinical" refers to the actual diagnosis and treatment of patients.

For weeks, Marcus had been checking the Centers for Disease Control and Prevention (CDC) website in the evenings after work. The virus, referred to as SARS-CoV-2, was spreading across the globe and seemed to be taking hold in the United States, with cases of the associated respiratory disease COVID-19 increasing daily. The World Health Organization (WHO) had officially classified the outbreak as a pandemic, and large numbers of cases were being reported in the state where Marcus lived.

When he arrived for work that morning, the laboratory director, Dr. Elaine Cordozo, called an emergency staff meeting. When everyone had gathered in the conference room, she made an announcement. “Working with our state’s health department, we have been given the important job of serving as a sample processing and testing laboratory for SARS-CoV-2 infection. We will begin preparations for conducting this testing immediately and will commence testing of patient samples as soon as possible.”

The next several days were a blur for Marcus and his colleagues. They worked quickly to get the laboratory testing set up. Finally, the day arrived that they would commence with SARS-CoV-2 diagnostic testing.

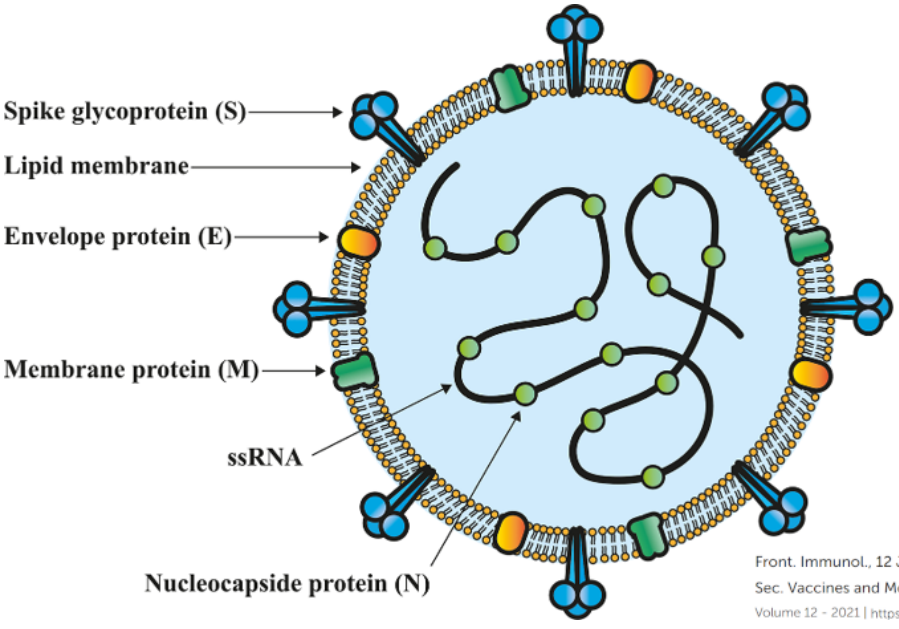
How does SARS-CoV-2 testing work? How do healthcare professionals make a diagnosis of SARS-CoV-2 infection?

In this TBL, we will follow Marcus through the process. Before we begin, read the following background information about the virus and the diagnostic procedure.

How coronavirus hijacks your cells

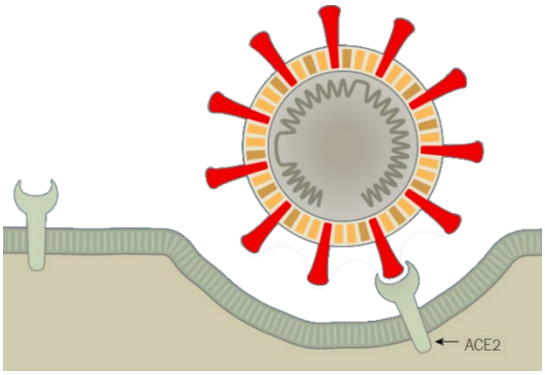
Besides SARS-CoV-2, at least six other types of coronavirus are known to infect humans, with some causing the common cold and two causing outbreaks: SARS and MERS.

The coronavirus is named after the crownlike spikes that protrude from its surface. The virus is enveloped in a bubble of oily lipid molecules, which falls apart on contact with soap.

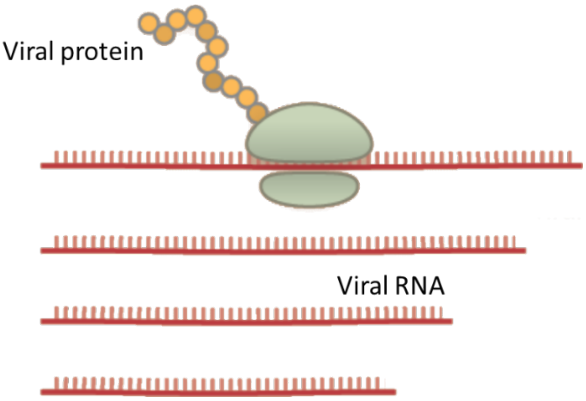


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Sec. Vaccines and Molecular Therapeutics
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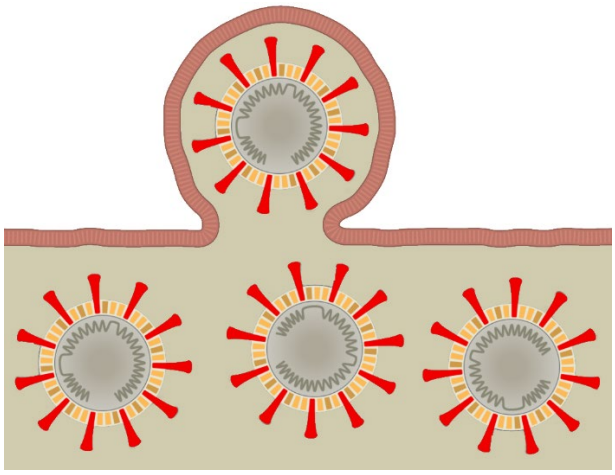
The virus enters the body through the nose, mouth or eyes, then attaches to cells in the airway that produce a protein called ACE2. The virus infects the cell by fusing its oily membrane with the membrane of the cell. Once inside, the coronavirus releases its genetic material in the form of RNA.



The virus's genome is less than 30,000 bases long (ours is over 3 billion). Scientists have identified genes for as many as 29 proteins, which carry out a range of jobs from making copies of the coronavirus to suppressing the body's immune responses. The infected cell reads the RNA and begins making proteins that will keep the immune system at bay and help assemble new copies of the virus.



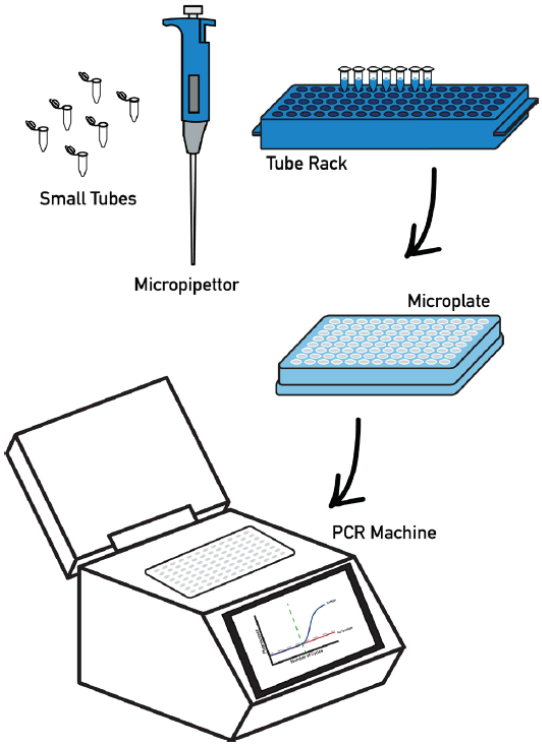
As the infection progresses, the machinery of the cell begins to churn out new spikes and other proteins that will form more copies of the coronavirus. New copies of the virus are assembled and carried to the outer edges of the cell.



SARS-CoV-2 infection can be asymptomatic or cause symptoms such as fever, cough, and fatigue. In severe cases, the immune system can overreact and start attacking lung cells. The lungs become obstructed with fluid and dying cells, making it difficult to breathe. A small percentage of infections can lead to acute respiratory distress syndrome, and possibly death.

SARS-CoV-2 diagnostic test protocol

General equipment used for sample preparation



Let's take a closer look at the equipment used and at each stage of the protocol to better understand how a sample is processed and analyzed for SARS-CoV-2.

After individual patient samples arrive at the diagnostic lab, they are processed in small tubes stored on tube racks. Micropipettors are used to draw up liquid solutions used in the various steps of this protocol. Multiple samples can be handled in parallel together on microplates to save time. These plates contain 96 wells, allowing multiple samples and controls to be analyzed in parallel using a PCR machine.

Stages of the test protocol

Stage 1 – Nucleic Acid Extraction

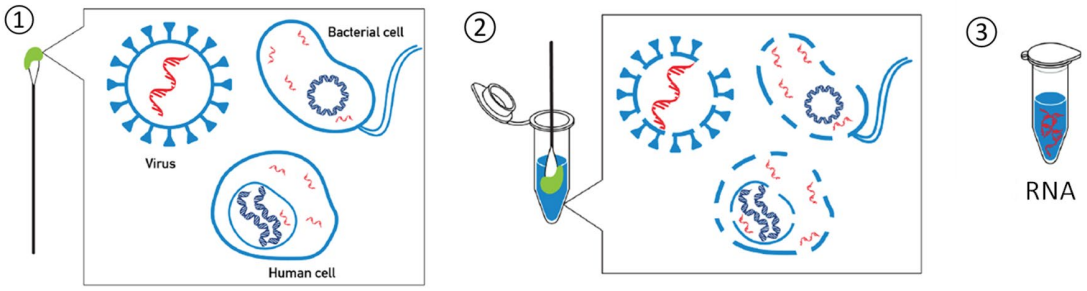
After the sample arrives in the diagnostic lab, the first step is to extract and purify nucleic acids from the patient's sample. These may include viral RNA as well as nucleic acids from human cells and other microorganisms. Substances that could interfere with the subsequent reactions are removed.

The process is as follows:

1. Obtain the patient sample.

2. Release nucleic acids from the viral particles and cells present in the sample.

3. Purify the nucleic acids and remove substances that could interfere with subsequent reactions.



Stage 2a Reverse Transcription (RT)

Once the RNA from the virus is isolated, it is ready to be amplified, detected, and analyzed. The first half of Stage 2 (Stage 2a) requires the viral RNA to be converted to complementary DNA copies, called cDNA for short. This process of making a DNA copy of RNA is called reverse transcription. This is accomplished by a special enzyme called reverse transcriptase.

The process is as follows:

1. The viral RNA isolated from Stage 1 is in a solution.

2. The solution contains short single-stranded DNA oligonucleotides, called primers, that bind to the RNA sequence.

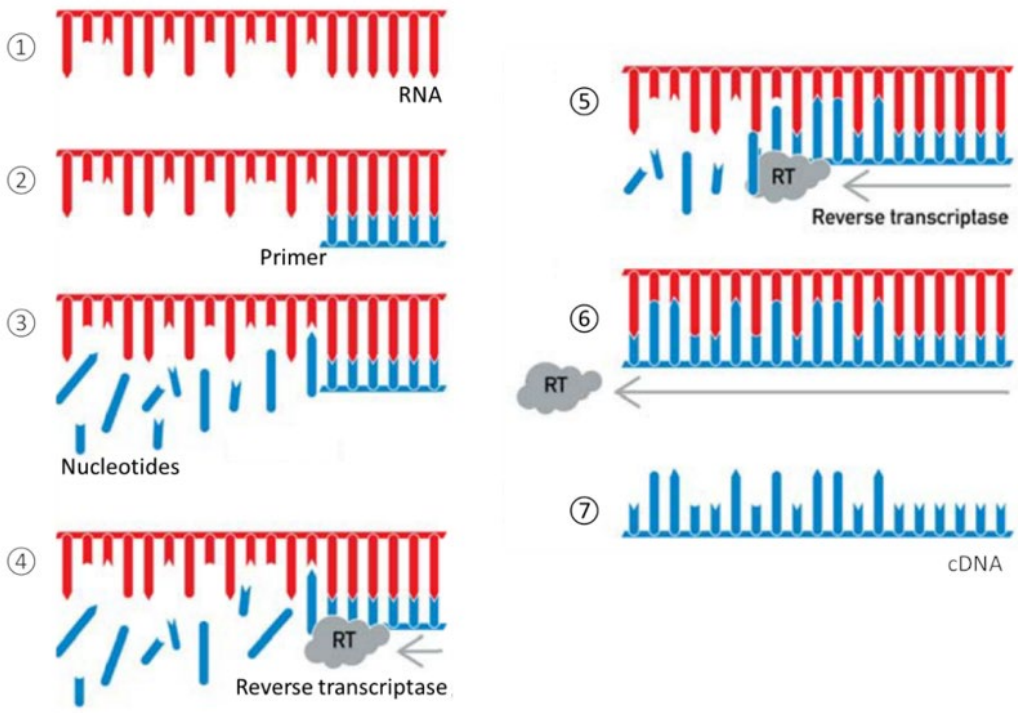
3. The solution also contains free DNA nucleotides (dATP, dCTP, dTTP, and dGTP).

4. The solution also contains the reverse transcriptase (RT) enzyme. The RT enzyme uses the primers to begin adding nucleotides complementary to the RNA.

5. The RT enzyme works along the RNA template, reading the RNA sequence and adding the complementary DNA nucleotides.

6. The RT enzyme makes “complementary” copies of the RNA present.

7. The cDNA is now ready for amplification by PCR.



Stage 2b Quantitative PCR

The second part of Stage 2 serves to amplify and detect the cDNA generated in Stage 2a using the PCR technique. Watch this video as an introduction to this technique.

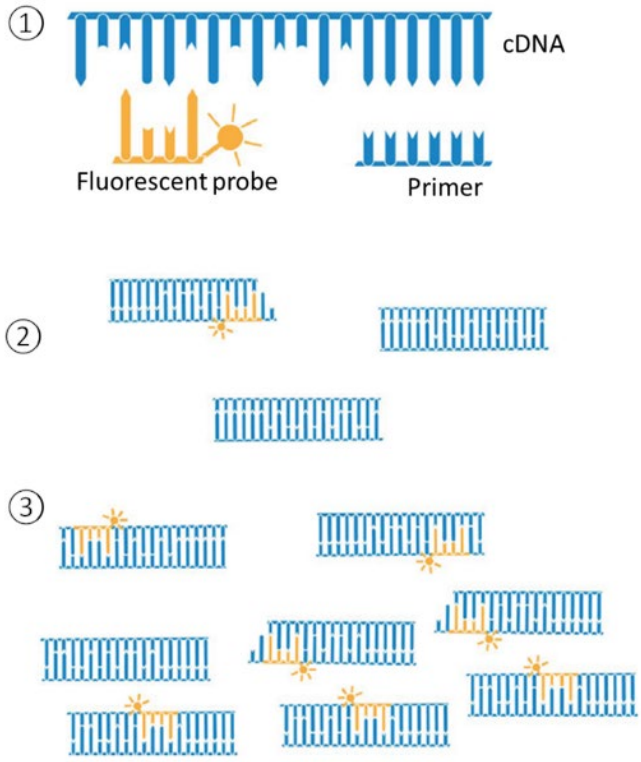
Like traditional PCR, quantitative PCR (qPCR) cycles through temperatures in order to amplify a small number of DNA copies into a large number. Unlike conventional endpoint PCR, qPCR monitors amplification as it occurs by measuring fluorescence. With appropriate standards, it can also estimate the initial amount of target nucleic acid.

The process is as follows:

1. The cDNA generated in Stage 2a is used as a template for the qPCR amplification. Primers find the target sequence to help the polymerase start to amplify and make more copies. Fluorescent probes are in the solution that will detect the target sequence as more copies are made during each cycle of the PCR.

2. As in a traditional PCR, the target sequence is amplified exponentially.

3. During DNA synthesis, the DNA polymerase cleaves probes bound to the target sequence, separating a fluorescent reporter from a quencher. This increases fluorescence as amplification proceeds. Samples containing more target RNA initially generally cross the fluorescence threshold after fewer cycles. Fluorescence is detected by a fluorometer inside the qPCR machine.



Stage 3 – Data Interpretation

The graph below shows the type of data collected from a quantitative PCR (qPCR) assay. Let's break down the components of the graph to analyze the results:

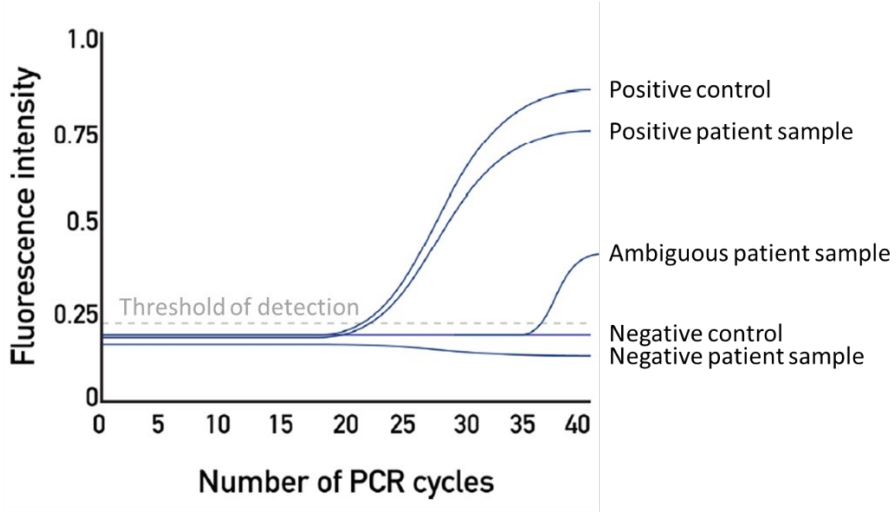
• Fluorescence (y axis) is tracked and reported for each sample across all cycles of the PCR assay (x axis).

• The threshold of detection (Line D) is the amount of fluorescence needed to distinguish true fluorescence signal from background signal from the assay.

• A positive control (Sample A) is included as a reference for what a virus-containing sample should look like; positive samples should create a high fluorescence signal, far above the threshold of detection.

• A negative control (Sample E) is included as a reference for what a sample without virus should look like; negative samples should yield a low fluorescence signal, below the threshold of detection.

• In this graph, three patient samples (Sample B, Sample C, and Sample F) illustrate different outcomes from this assay.



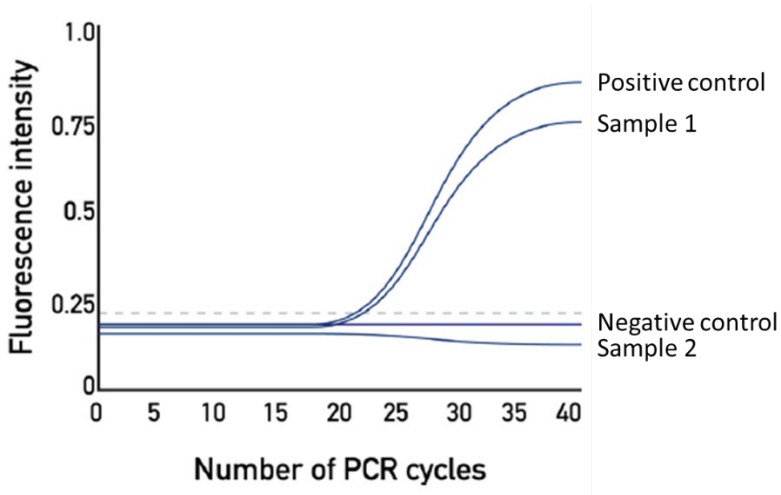
In summary

Let's look at two patient samples to summarize the laboratory process:

1. Nucleic acids are extracted and purified from the two samples that arrive at the lab.

2. The RNA is converted to cDNA via reverse transcription. The cDNA is amplified by qPCR, and fluorescence generated by cleavage of the probes is detected.

3. Data are graphed from the qPCR assay.



Sample 1 shows a high signal detected early in the cycle run, similar to the positive control. Sample 1 is reported as SARS-CoV-2 positive. Sample 2 shows a low signal below the threshold of detection, similar to the negative control. Sample 2 is reported as SARS-CoV-2 negative (assuming sample quality and successful processing).